Certified-Everlasting Quantum NIZK Proofs

Nikhil Pappu



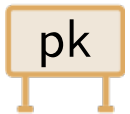
UC SANTA BARBARA

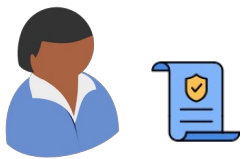
Certified-Deletion from No-Cloning

Encryption w/ Certified-Deletion [BI20,HMNY21,BK23]

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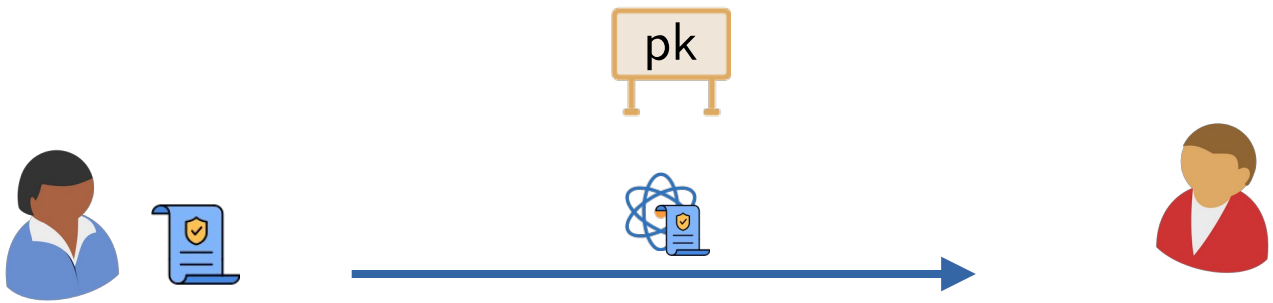
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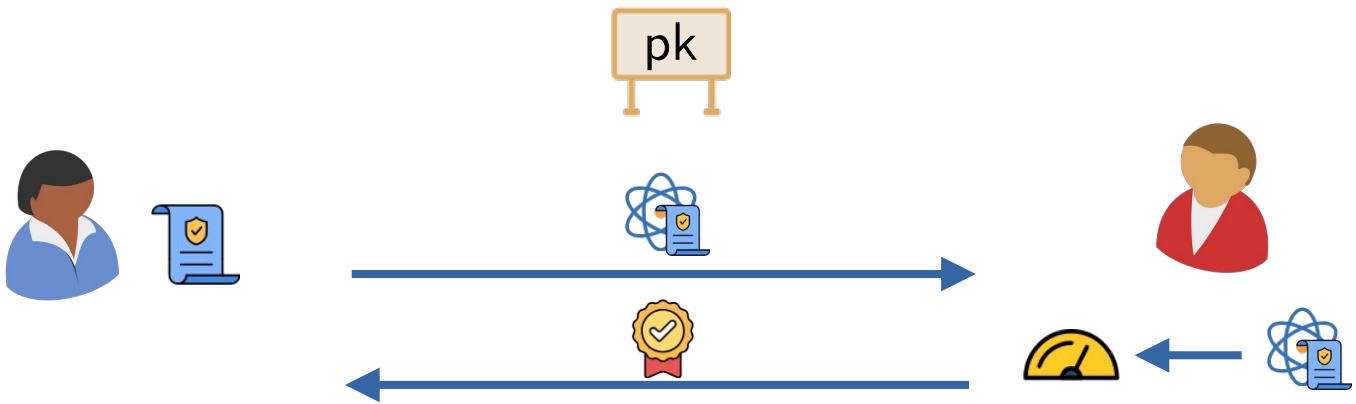
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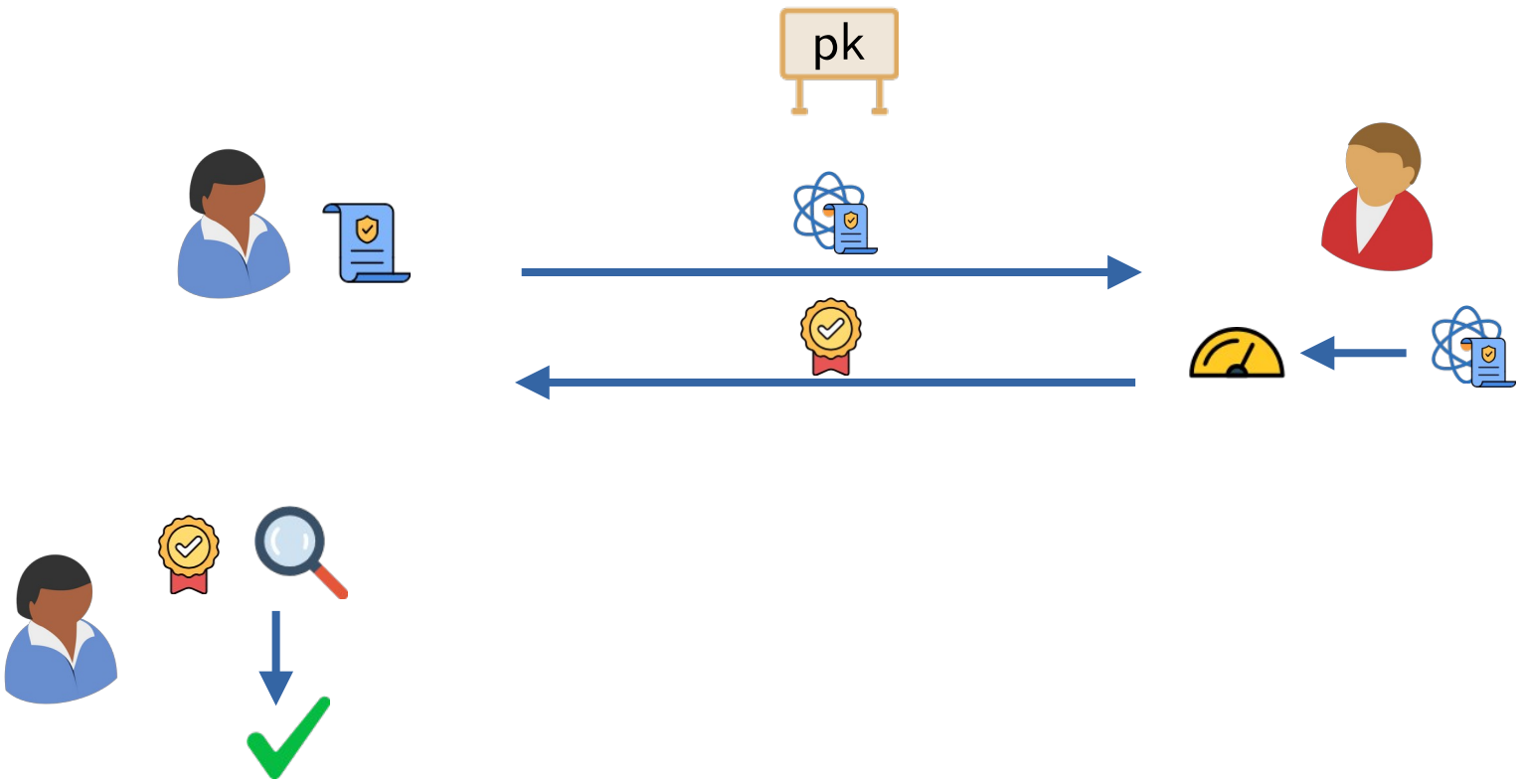
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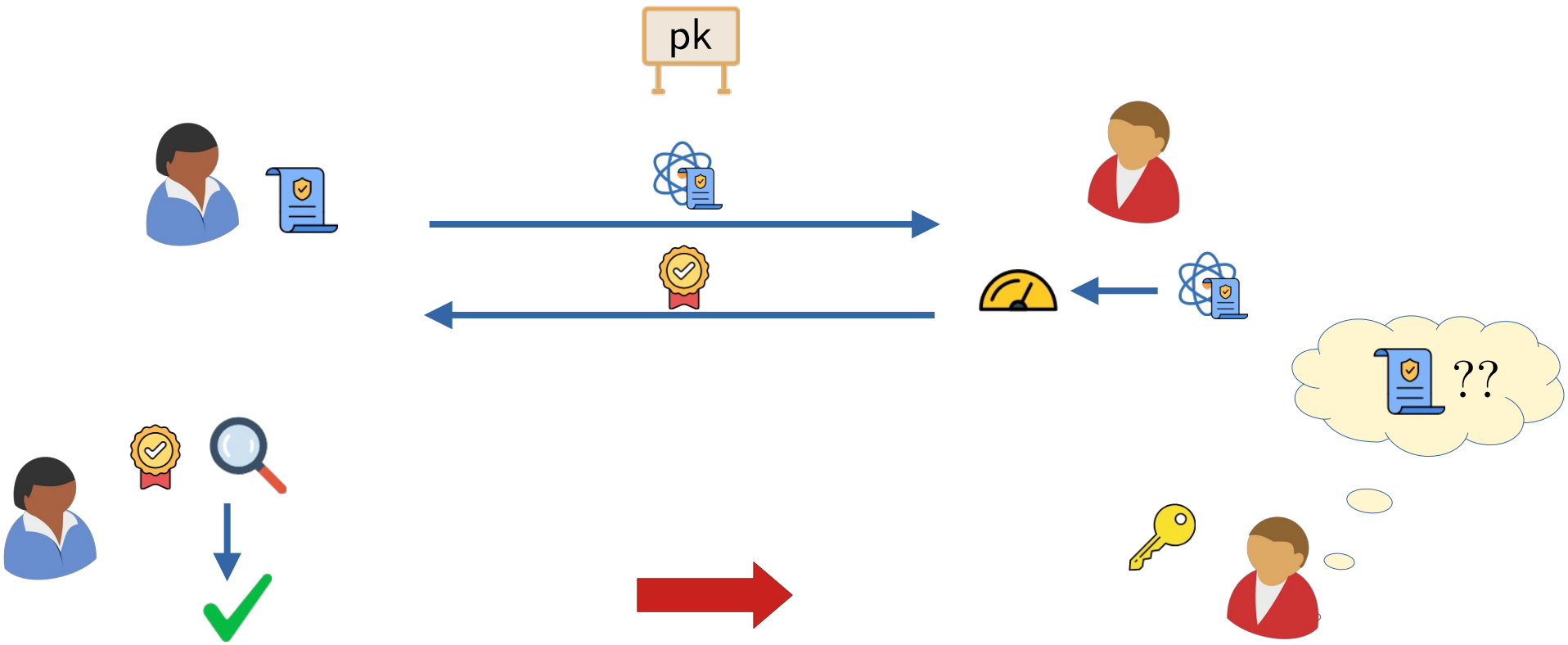
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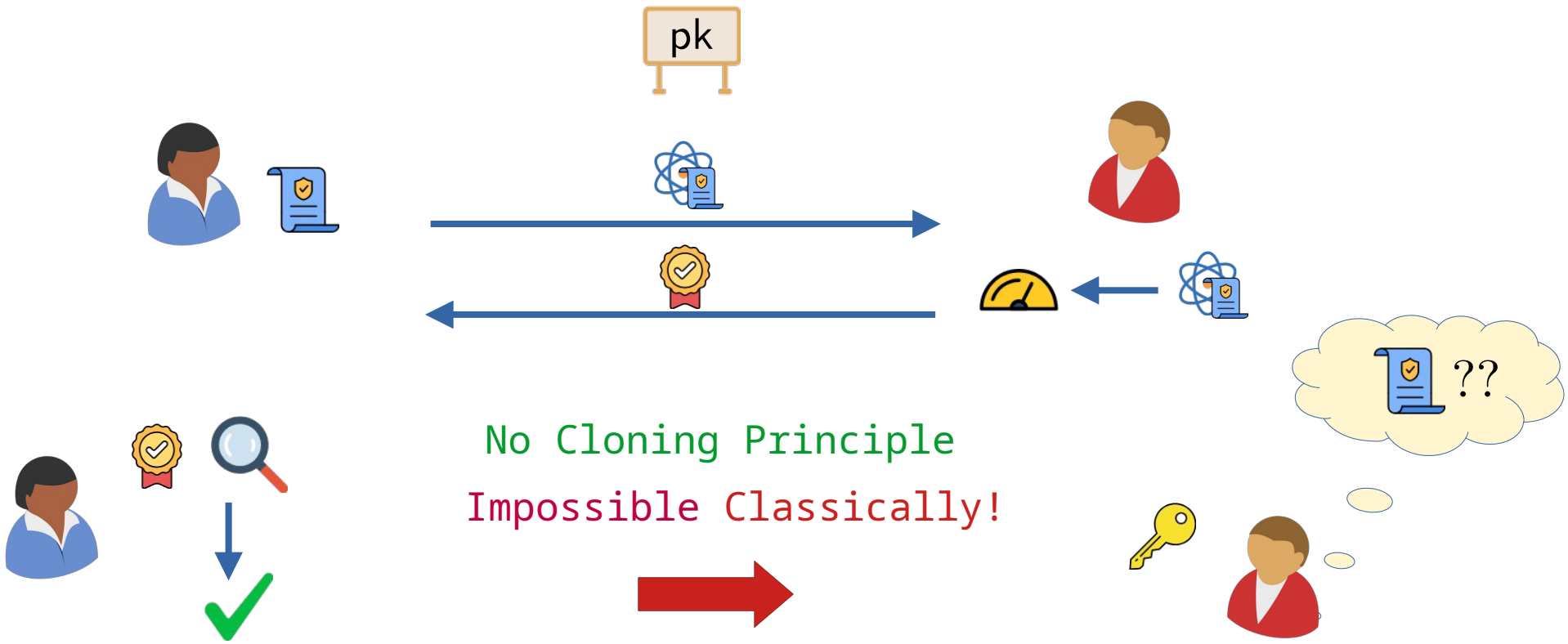
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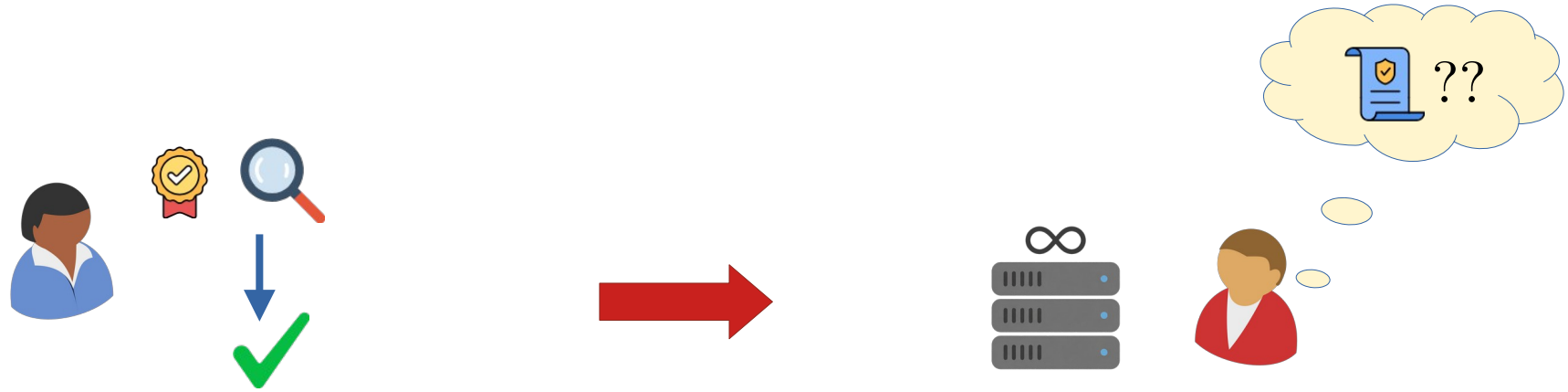
Certified-Everlasting Security

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Certified-Everlasting Security: Admits **Unbounded** Post-Deletion Bob

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Certified-Everlasting Literature

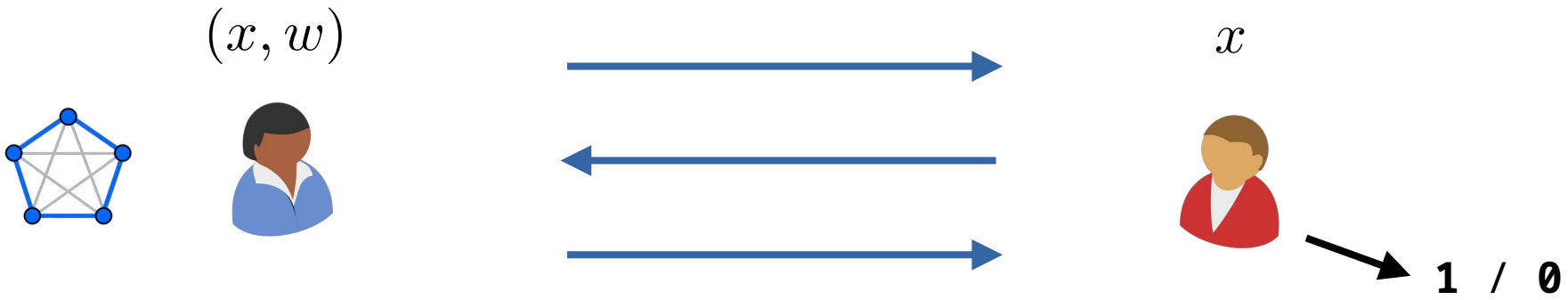
Work	Primitives	Assumptions	States	Public-Verifier
[BI20]	**SKE**	None	BB84	No
[HMNY22],[BK23]	**Com / ZK**	Com (OWF or PRS)	BB84	No
[BK23]	**PKE / FHE / WE**	PKE / FHE / WE	BB84	No
[HKM+24]	**GC / FE / CC-Obf**	OWF / iO / LWE	BB84-based	No

Certified-Everlasting Literature

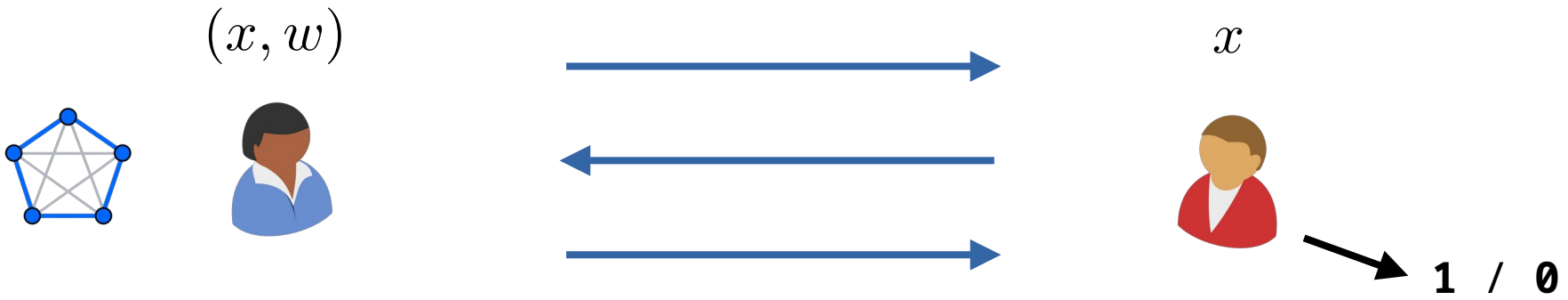
Work	Primitives	Assumptions	States	Public-Verifier
[BGK+24]	**diff-input-Obf**	iO + OWF	Coset	Yes
[BKP23]	**PKE / FHE**	LWE + SIS	Gaussian	Yes
[KNY23]	**Com, PKE, WE etc**	Primitive + OWFs	BB84-based	Yes
[BKM+23]	**Com, PKE, WE etc**	Primitive + OWFs	Two-Superposition	Yes

Our Focus: Zero-Knowledge Proofs

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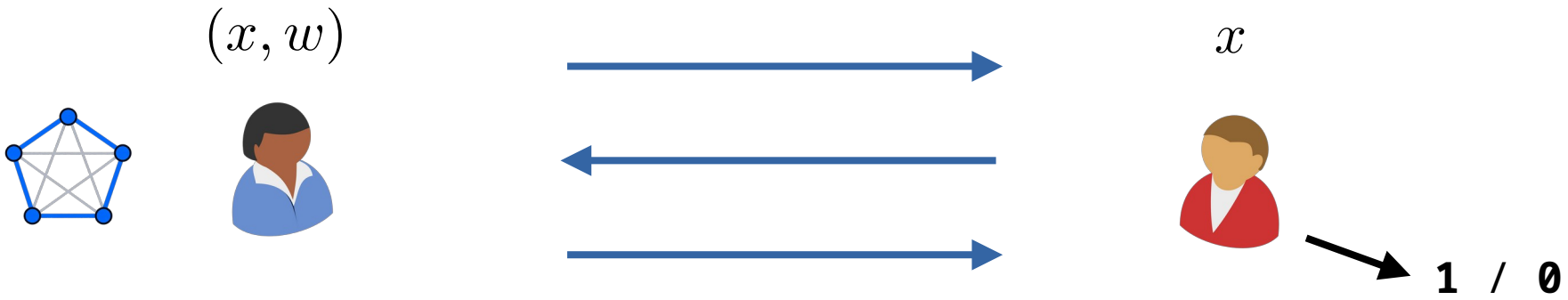


Our Focus: Zero-Knowledge Proofs



- **Statistical Soundness:** $x \notin L$ $\Rightarrow$ $\forall\, \mathsf{P}^* : \Pr[\langle \mathsf{P}^*, \mathsf{V}(x) \rangle = 1] \leq \mathsf{negl}(\lambda)$

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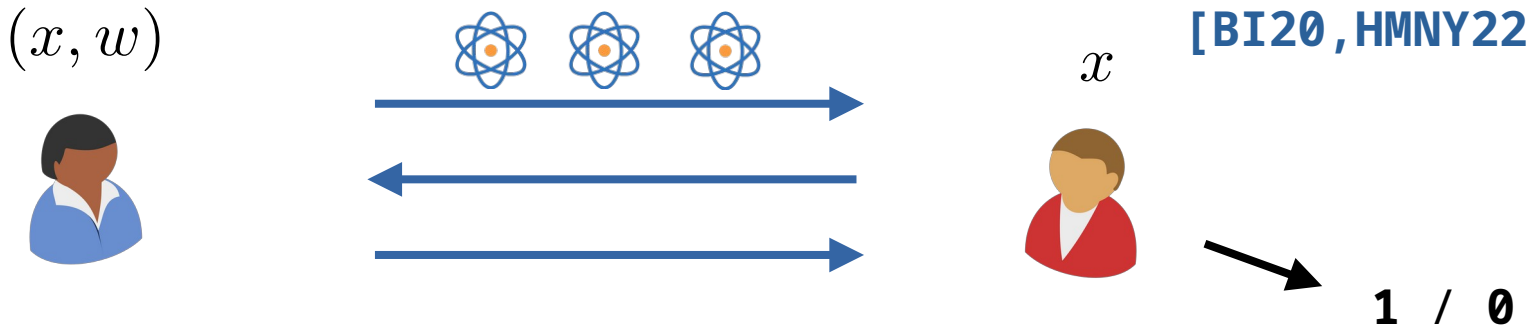
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⚠ **[For87]: Both Statistical is impossible for non-trivial languages.**

Quantum-Enabled Possibility

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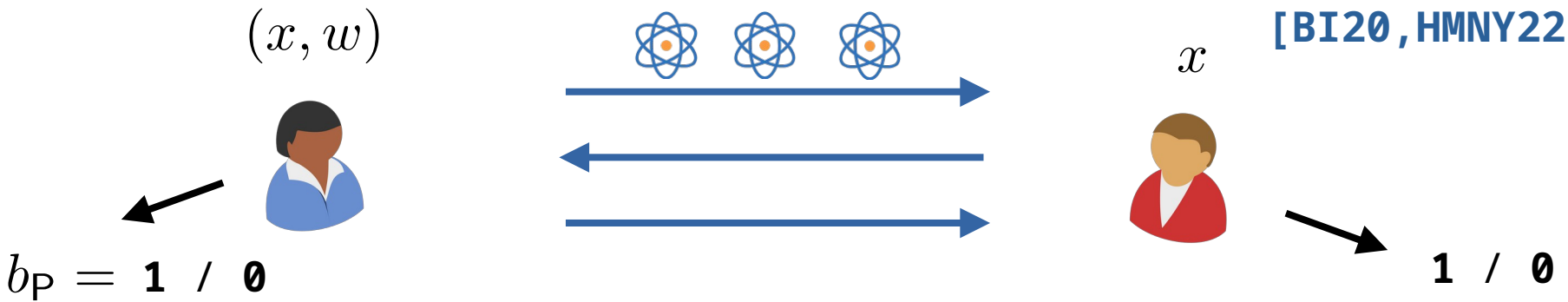
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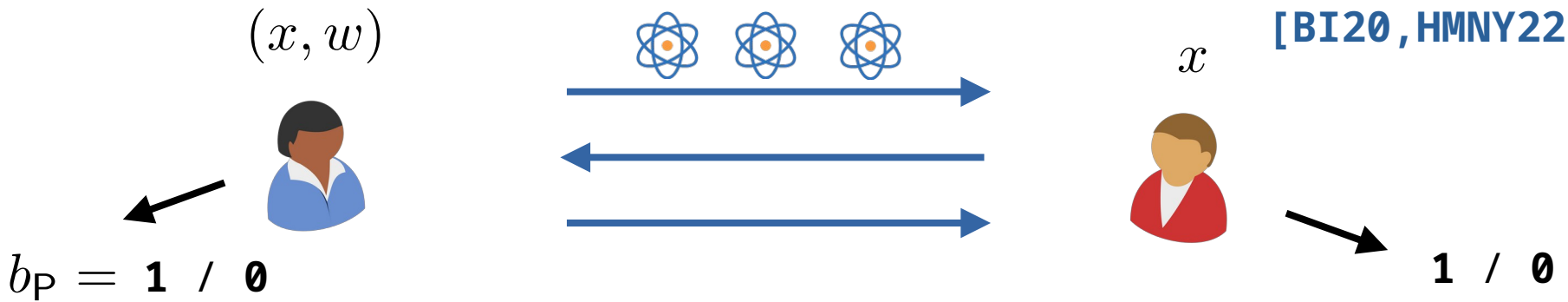
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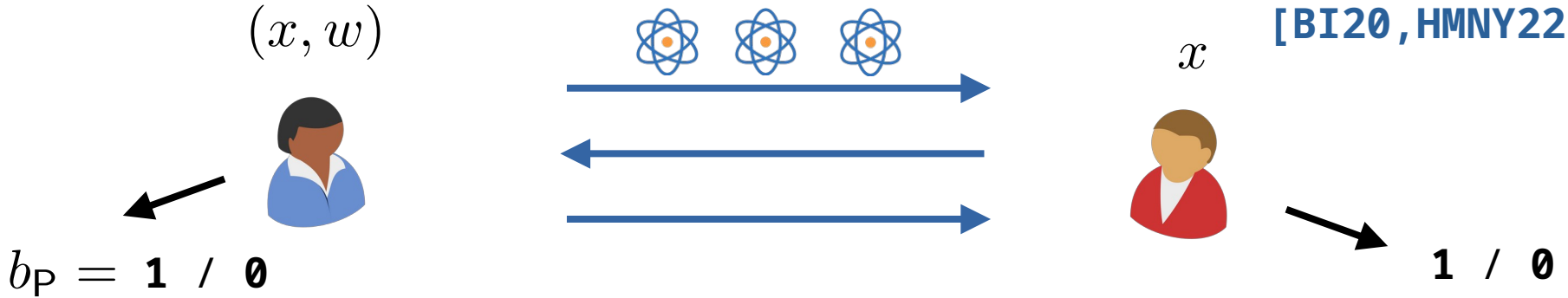
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- **Standard Notions:** Statistical Soundness, Computational ZK

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Quantum-Enabled Possibility



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$$\mathsf{Sim}^{\mathsf{V}^*}(x) \approx_s \mathsf{EV}(b_{\mathsf{P}}, \rho_{\mathsf{V}^*})$$

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Round Complexity?

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Statistically-Binding Commitments

[HMNY22,BK23]



Poly-round protocols

Certified-Everlasting ZK (CE-ZK)
Proofs for **NP** (even QMA)

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Poly-round protocols

Certified-Everlasting ZK (CE-ZK)
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Non-Interactive ??

NIZKs w/ Certified-Everlasting Security

Motivation:

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- **Building Block:** CCA-Secure PKE [NY90], Round-Efficient MPC [AJL+12]

NIZKs w/ Certified-Everlasting Security

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NIZKs w/ Certified-Everlasting Security

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- **Limited Reliance** on Hardness Assumptions like LWE.
- **Theoretical Interest:** Several Primitives w/ Certified-Everlasting Security Exist. Eg: Garbled Circuits, PKE, Commitments, ZK etc.

Our Contributions

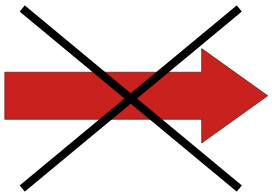
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2) **Barrier:** Interactive CE-ZK



Fiat-Shamir

CE-NIZK

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2) **Barrier:** Interactive CE-ZK ⟶ (Fiat-Shamir) ⟶ CE-NIZK

3) LWE ⟶ [WWW25] ⟶ Quantum-Secure NIZK ⟶ (Non-BB) ⟶ CE-NIZK

CRS Model

Our Contributions

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Similar barrier applies to **Hidden-Bits Model** approaches to NIZK…

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But **not** with **Pre-Shared Entanglement** between prover and verifier!

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4) 

Our Contributions

Similar barrier applies to **Hidden-Bits Model** approaches to NIZK…

But **not** with **Pre-Shared Entanglement** between prover and verifier!

4) **LWE** → **[WWW25]** → **Hidden Bits Generator** → **CE-NIZK**

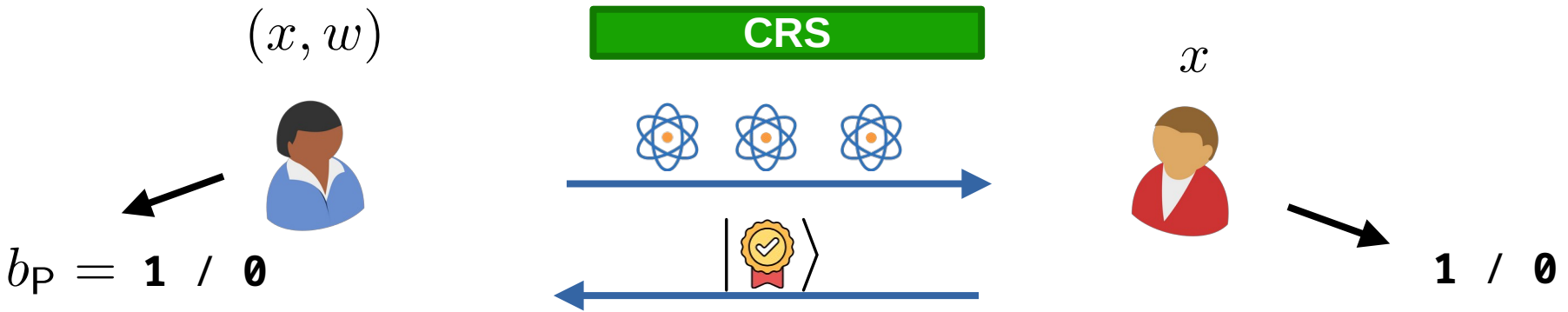
Shared EPR Model

Needs Only **Single-Qubit** Operations (EPR Measurements)

Certified-Everlasting NIZK

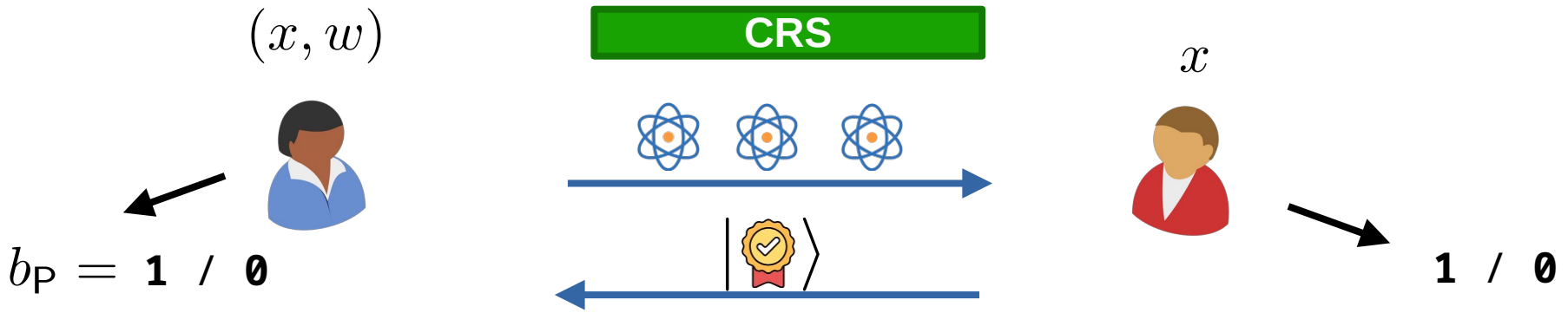
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Certified-Everlasting NIZK



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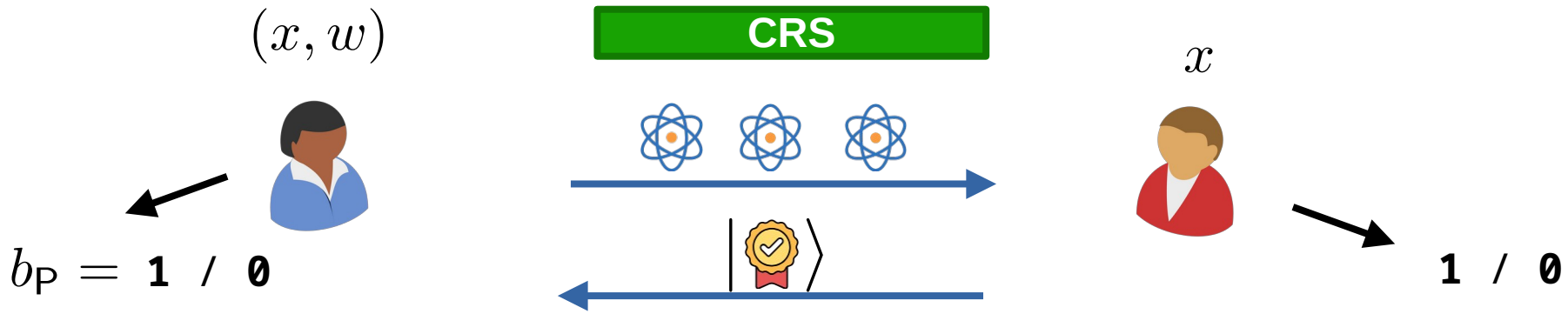
Certified-Everlasting NIZK



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Certified-Everlasting NIZK

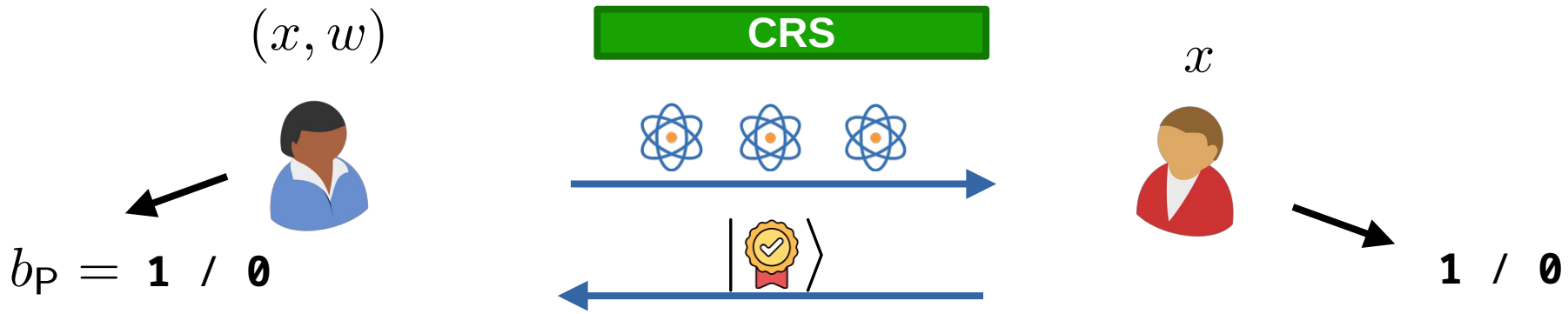


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$$\mathsf{EV}(b, \rho) := \begin{cases} (1, \rho) & \text{if } b = 1, \\ (0, \bot) & \text{if } b = 0. \end{cases}$$

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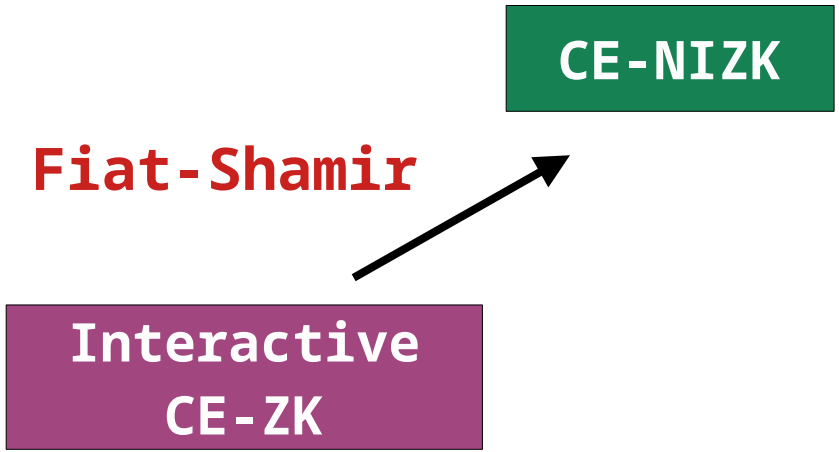
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Simulator can choose the CRS.

Barrier Intuition

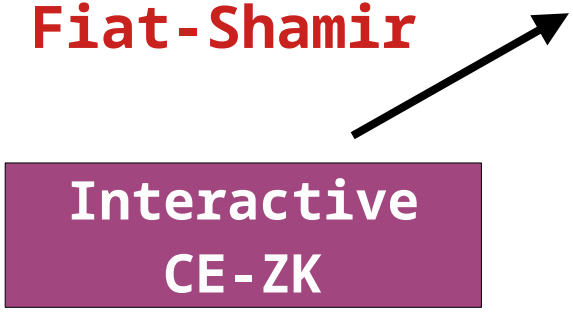
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CE-NIZK

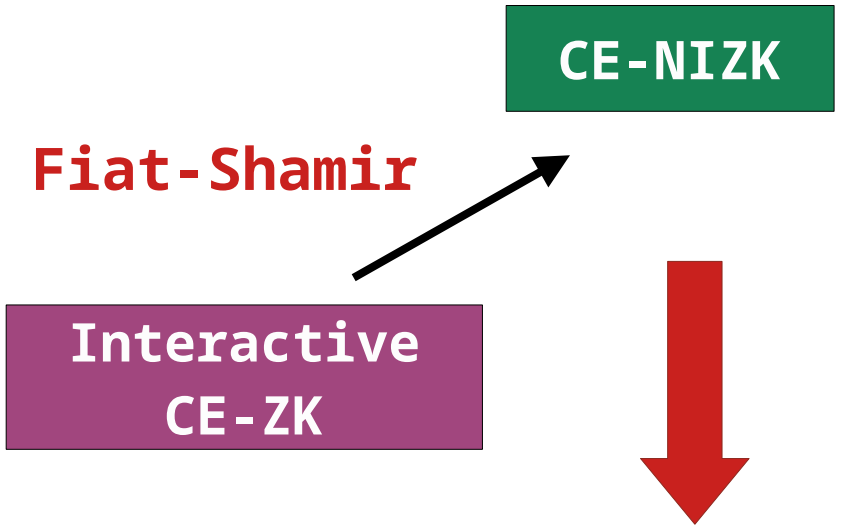
Fiat-Shamir





Prover can perform **deletion by themself** before sending the proof!

Barrier Intuition



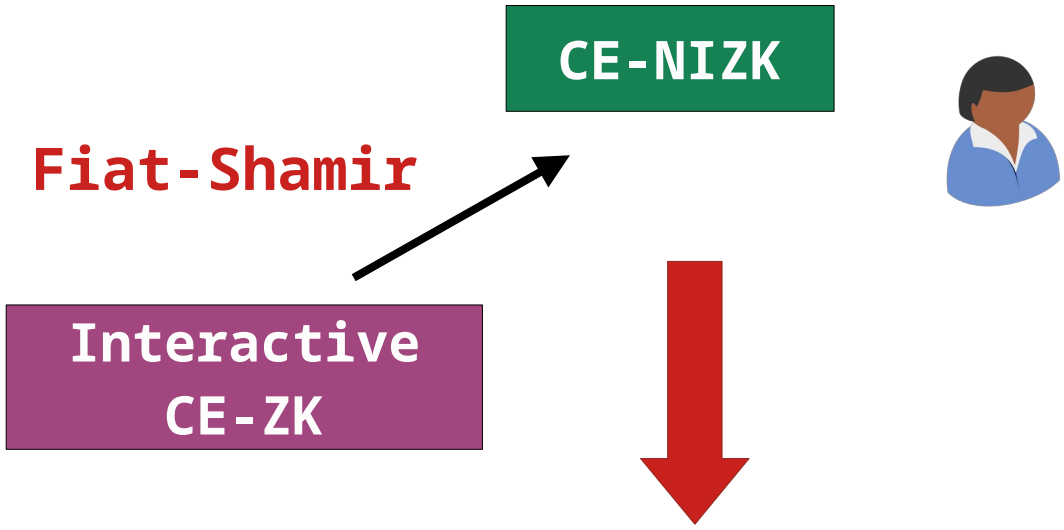


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NIZK with:

- **Statistical Soundness**
- **Statistical ZK**

Barrier Intuition



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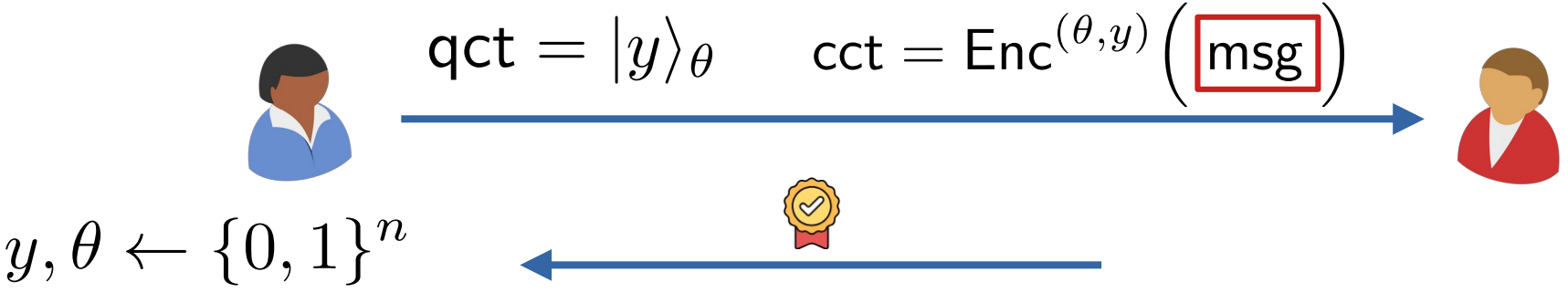
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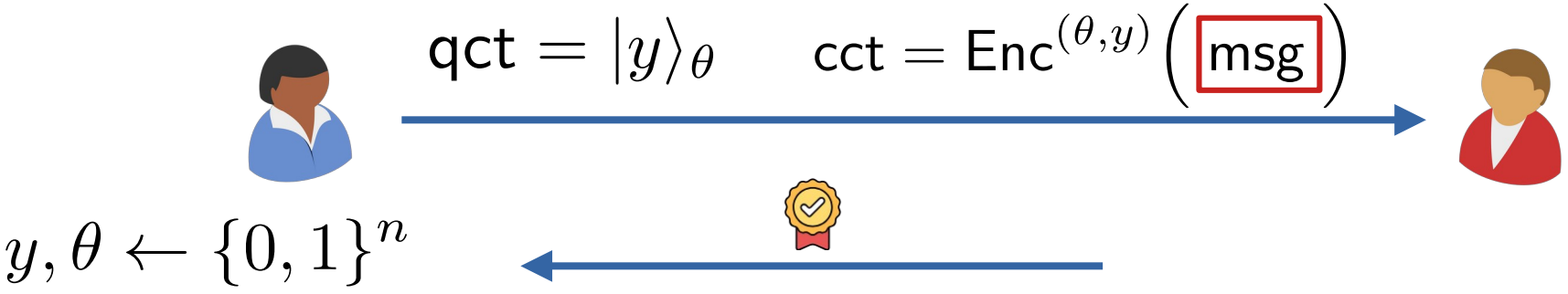
Does not exist for non-trivial languages [PS05].

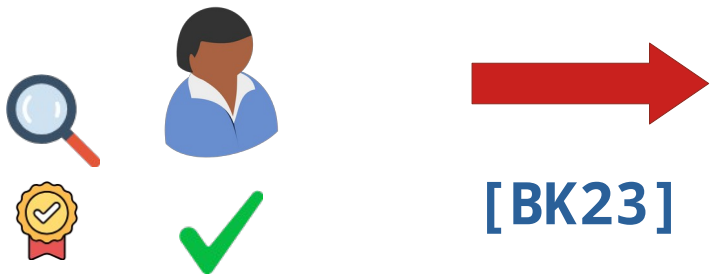
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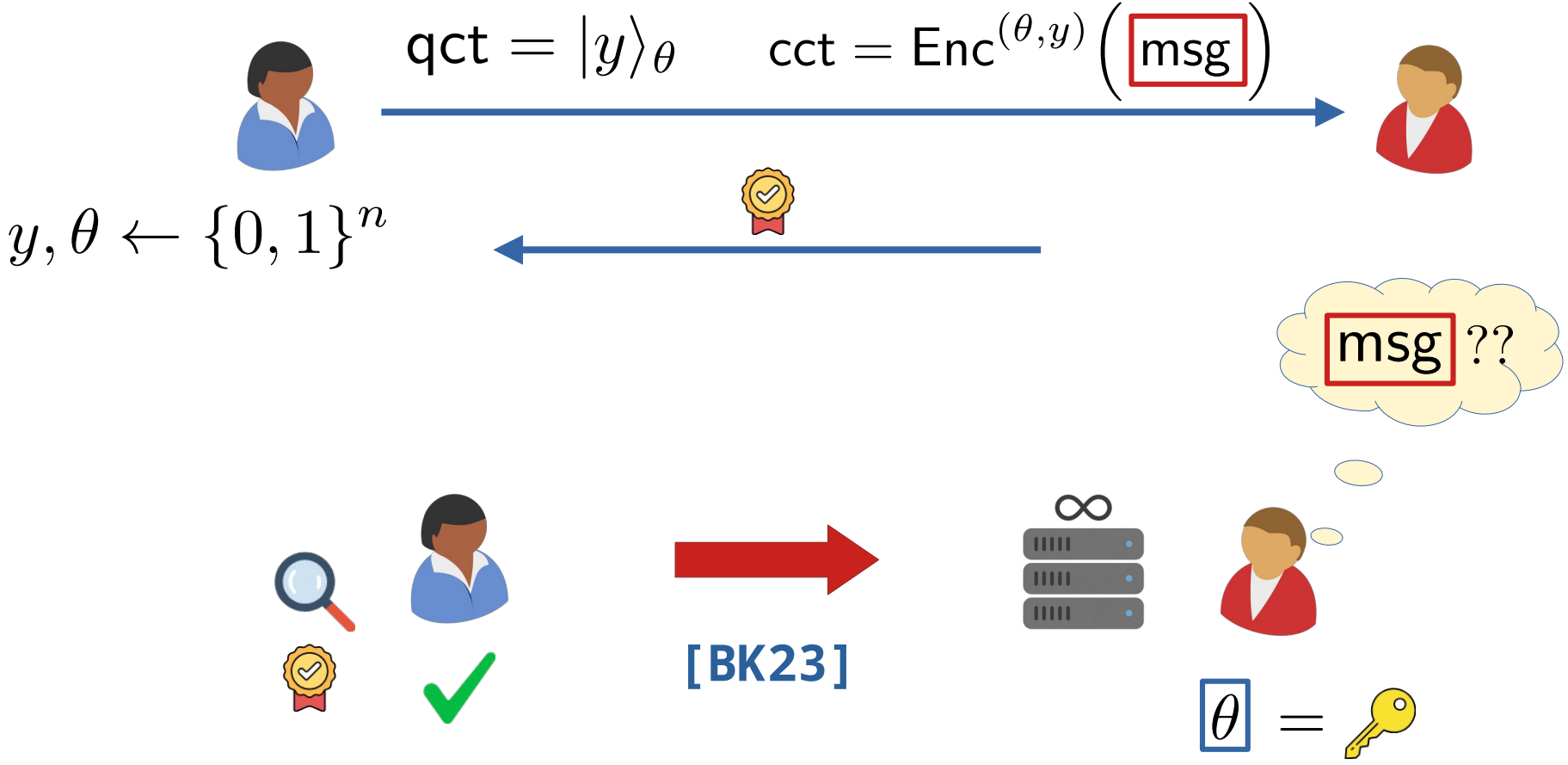


Encryption w/ Certified-Deletion





Encryption w/ Certified-Deletion



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$$\texttt{qs-NIZK} = \mathsf{NIZK}.(\mathsf{Setup}, \mathsf{P}, \mathsf{V}, \mathsf{Sim})$$

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$|y\rangle_\theta = \sum_z \alpha_z |z\rangle$

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$$\mathsf{cct}_0 = \mathsf{Enc}_0^{(\theta,y)}\left(\boxed{\pi_{\mathsf{in}}}\right) \qquad \mathsf{cct}_1 = \mathsf{Enc}_1^{(\theta,y)}\left(\boxed{0^\ell}\right)$$

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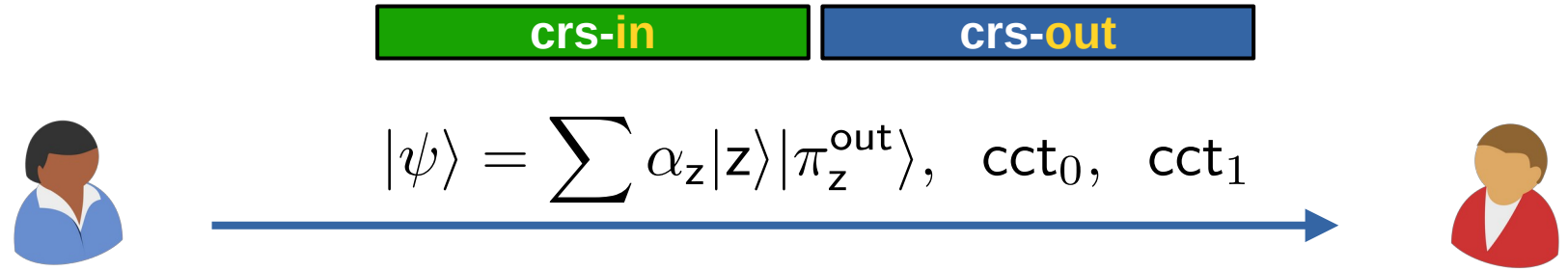


$\mathsf{cct}_0 = \mathsf{Enc}_0^{(\theta,y)}(\pi_{\mathsf{in}})$ $\quad \mathsf{cct}_1 = \mathsf{Enc}_1^{(\theta,y)}(0^\ell)$

Proves wrt crs-in **that:** $R_L(x, w) = 1$

Our Construction

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Our Construction

crs-in | crs-out



$$|\psi\rangle = \sum \alpha_z |z\rangle |\pi_z^{out}\rangle, \quad cct_0, \quad cct_1$$





Verifies π_z^{out} in superposition.

Our Construction

crs-in | crs-out



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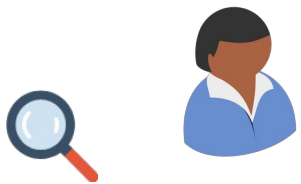


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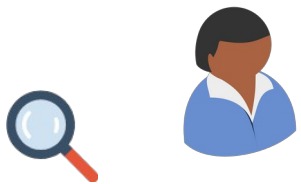
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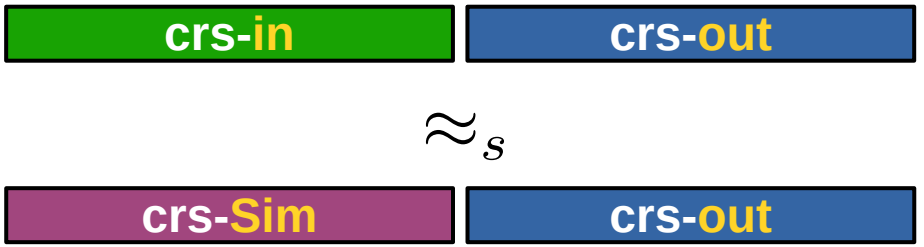
Verifies π_z^{out} **in superposition.**

π_z^{out} : **Proves wrt crs-out and using witness** $\omega_{\text{out}} = \theta$ **that at least One Ciphertext decrypts to an accepting NIZK proof. Note: Each z can decrypt** cct_i **(given** θ**).**

Proof Sketch

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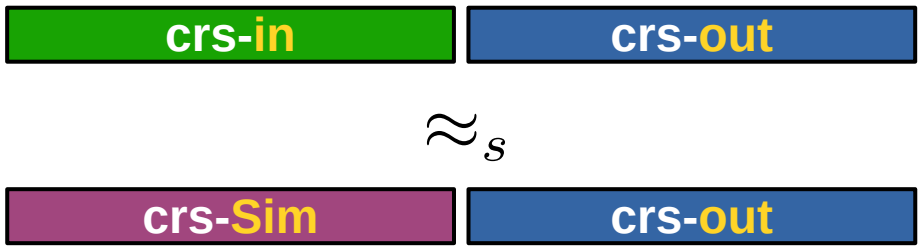
$\mathsf{NIZK.Sim}(x) \rightarrow (\boxed{\text{crs-Sim}}, \pi_{\mathsf{Sim}})$



Statistical CRS Indistinguishability

Proof Sketch

$\mathsf{NIZK.Sim}(x) \to (\boxed{\text{crs-Sim}}, \pi_{\mathsf{Sim}})$



Statistical CRS Indistinguishability

$$\mathsf{Enc}_0^{(\theta,y)}\left(\boxed{\pi_{\mathsf{in}}}\right) \qquad \mathsf{Enc}_1^{(\theta,y)}\left(\boxed{0^\ell}\right)$$

$$\approx_s$$

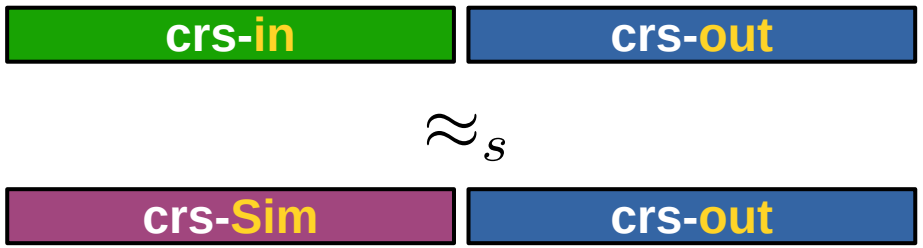
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Hybrids output $\perp$ when deletion verification fails.

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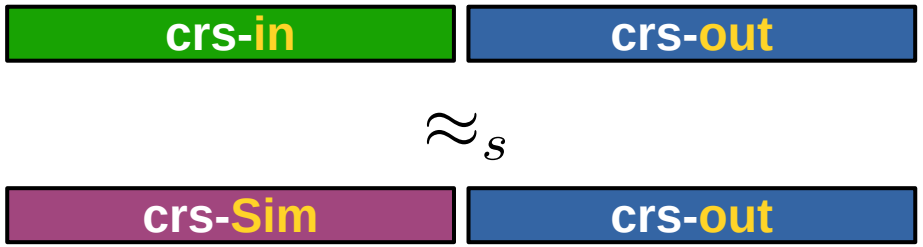
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Hybrids output $\perp$ when deletion verification fails.

Simulatable w/o witness.

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$$\mathsf{cct}_0 = \mathsf{Enc}_0^{(\theta,y)}\left(\boxed{\pi_{\mathsf{in}}}\right) \qquad \mathsf{cct}_1 = \mathsf{Enc}_1^{(\theta,y)}\left(\boxed{0^\ell}\right)$$

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- **Transitions** are argued by **reducing** to the **deletion-theorem** for BB84 states by **[BK23]**.

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- **Transitions** are argued by **reducing** to the **deletion-theorem** for BB84 states by **[BK23]**.
- To **invoke** the deletion-theorem, we **rely on proofs** $\boxed{\pi_{\mathsf{z}}^{\mathsf{out}}}$ in superposition to be **computationally** simulatable **w/o** $\boxed{\omega_{\mathsf{out}} = \theta}$.

Open Problems

- **Classical** certificates
- **Publicly-Verifiable** Deletion
- Non-Interactive **Witness-Indistinguishability** (in Plain Model)
- Other Primitives with **Simulation-Based Deletion** Guarantees

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Thank You!